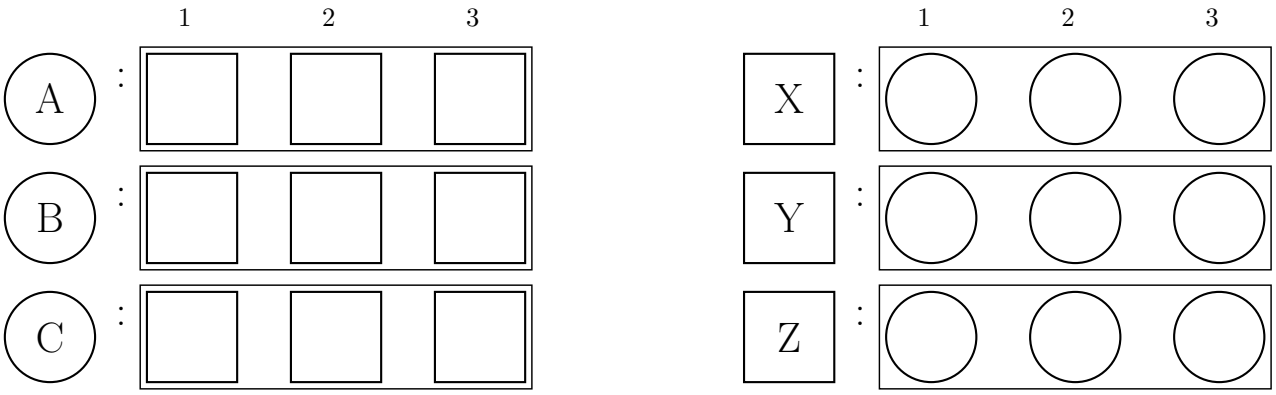
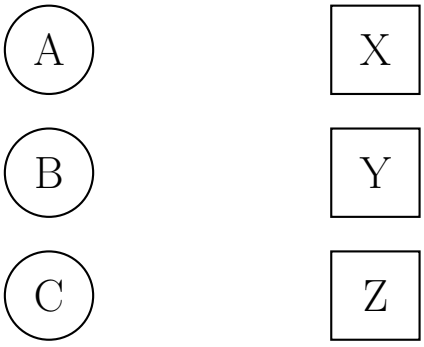
Pair each circle with one square, using every letter once. Each list runs from first choice to last, with no ties. A pairing is stable if no two letters in different pairs both prefer one another to their current partners.

Problem 3: Change the rule: a receiver must keep the first asker and reject every later asker. Choose complete lists for A, B, C and X, Y, Z that make these changed rules end with an unstable pairing. Let A, B, C ask; record the requests and two letters in different pairs that both prefer each other.





Problem 4: Four letters on each side follow the asking rules from Problem 2. Give a number of requests that can never be exceeded, and explain why the rules must finish with everyone paired. Would your argument work with ten letters on each side?

Problem 5: When the asking rules from Problem 2 finish, could two letters in different pairs both prefer one another to their final partners? Explain why your answer holds for every set of complete lists with no ties and equal-sized sides.

Problem 6: Choose complete lists for A, B, C and X, Y, Z and a stable pairing in which as many of the six letters as possible get their last choice. Explain why a larger number is impossible.

